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Oefeningenreeks 1C nr. 31.3

$$\left(-\frac{1}{\sqrt{3}} - i\right)^8$$

$$\begin{cases} r = \sqrt{\left(-\frac{1}{\sqrt{3}}\right)^2 + (-1)^2} \\ \tan\theta = \dfrac{-1}{-\frac{1}{\sqrt{3}}} \end{cases} \Rightarrow \begin{cases} r = \dfrac{2\sqrt{3}}{3} \\ \theta = \dfrac{\pi}{3} + \pi \end{cases} \Rightarrow \begin{cases} r = \dfrac{2\sqrt{3}}{3} \\ \theta = \dfrac{4}{3}\pi \end{cases}$$

$$\Rightarrow \left(\frac{2\sqrt{3}}{3} \operatorname{cis} \frac{4}{3}\pi\right)^8$$

$$= \left(\frac{2\sqrt{3}}{3}\right)^8 \operatorname{cis} \frac{4\pi}{3} \cdot 8$$

$$= \frac{256}{81}\left(\cos\frac{32\pi}{3} + i \cdot \sin\frac{32\pi}{3}\right)$$

$$= \frac{256}{81}\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$

$$= -\frac{128}{81} + \frac{128\sqrt{3}}{81}i$$

References